Project Management Plan

for

Project THEIA

Version 1.9.1 draft 4

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Washington State University

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Revision History

Name	Date	Reason for Changes	Version
Preliminary Plan Draft 1	9/17/2026	initial draft	1.0 draft 1
Preliminary Plan Draft 2	9/18/2026	Revised draft; filled out sections	1.5 draft 2
Preliminary Plan Draft 3	9/19/2026	Filled out remaining sections	1.9 draft 3
Preliminary Plan Draft 4	10/15/2026	Fixed parts based on feedback	1.9.1 draft 4

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1. Overview

Project THEIA is a mobile phone application that helps the visually impaired and blind people navigate indoors. Users are able to find where they are, choose a destination and follow a route, be able to turn and stop in the right places, avoid obstacles, and get help if issues arise.

1.1. Project Purpose, Objectives, and Success Criteria

1.1.1 Purpose and Business Needs

The purpose of this application is to assist the blind and visually impaired navigate indoors and addresses the business need for an intuitive mobile solution providing navigation within or to another building.

The app will

- Provide information about their current whereabouts.
- Provide locations to go to.
- Direct them through a planned route.
- Detect obstacles and help avoid collisions.
- In case of emergencies, have a hotline they can call for help.

1.1.2 Quantitative and Measurable Business Objectives

The app should

- Behave in a safe manner, and without any incident(s).
- Behave in an efficient manner, creating a route is comfortable and familiar.
- Be usable without vision, meaning key functions of the application such as navigation and knowing current location should be accessible through voice commands or tactile input.

1.1.3 Success Criteria for Stakeholders

The stakeholders of this application will be the blind and visually impaired, caretakers, police, and even the app admin. The primary stakeholders are the blind and visually impaired, where using the application they can reliably and consistently make it to their destination alone safely. The secondary stakeholders will be the caretakers,who can configure the application for the user and be able to respond to emergency notifications should there be any issues with the user. Another secondary stakeholder is the police, who will be able to easily locate a distressed user. The last stakeholder is the admin, who will be able to update the building layout in case of changes to the facility or closed off sections of buildings.

Success criteria for the app will include how efficient the routes are, how few incidents are caused by app direction, how easy to understand and navigate the app controls are, and how adaptable the app is to updates or expansion of routes.

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1.2. Project Deliverables

The project will be delivered in 2 phases, with **Phase I** focusing on requirement gathering and prototyping. The key deliverables for this phase will include:

- Requirement Documentation: A comprehensive specification of the app's functional and non-functional requirements, utilizing the WRS template.
- Prototyping and User Manual: A simple mockup of the app to demonstrate core functionality and UX with a comprehensive user manual for ease of understanding.
- Project Plan: A detailed plan for Phase I, including team organization, roles, and team meetings.

Deliverable	Delivery Date	Delivery Method	Comments
Project Phase 1: Preliminary Plan Submission	09/20/2026	Canvas & Repo	The preliminary plan document.
Project Phase 1: Peer Review	10/12/2026	Canvas	Peer reviews for all teammates.
Project Phase 1: Final Submission	10/18/2026	Canvas & Repo	WRS Revised & Updated Phase 1 Plan Slides

1.3. Assumptions, Dependencies, and Constraints

1.3.1 Assumptions

AS-1: Users have access to a smartphone with sensors such as camera, accelerometer, gyroscope, microphone access given to the app with accessibility features enabled.
AS-2: Buildings with prior approval for app navigation will have or are willing to provide the floor plan or map.
AS-3: Users are familiar with built in accessibility features such as TTS, as the app will be relying on these systems.
AS-4: Users fall within the category of blind or visually impaired.

1.3.2 Dependencies

DE-1: Access to accurate building maps and/or positioning aids.
DE-2: Integration with built-in phone accessibility APIs and voice assistant features.
DE-3: User testing and feedback for safety verification and ease of use.

1.3.3 Constraints

CO-1: Indoor GPS is finicky and inaccurate.
CO-2: Smartphone battery and/or lack of processing power to utilize sensors.
CO-3: Safety and usability relies too heavily on user feedback and tests.
CO-4: User smartphones may not have specific sensors, and need fallback.

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1.3.4 TBD

TBD-1: How to define destination locations.
TBD-2: User customization (if needed).
TBD-3: Emergency protocol undefined.
TBD-4: Risk Handling undefined.
TBD-5: Define how to take user feedback.

1.4. References

- Project Management Plan Template (Wiegers, 2013).
- Requirements Review Checklist (Wiegers & Seilevel, 2013).
- Agile Development Models - Software Engineering (GeeksForGeeks, 2026).
- https://github.com/Fleronin/theia-app

1.5. Definitions and Acronyms

- THEIA - The working app name.
- WRS - Wieger's Requirements Specification, a template used to document the app's functional and nonfunctional requirements.
- TTS - Text to Speech.
- GPS - Global Positioning System.
- API - Application Programming Interface.
- UX - User Experience.
- GUI - Graphic User Interface.
- AS / DE / CO / TBD - Acronyms for Assumption, Dependency, Constraints, and To-Be-Determined.
- YOLO - You Only Look Once, an object detection system.

2. Project Organization

2.1. Process Model

The team will be following an agile development process, with consistent reiterations of the product depending on user feedback, test results, or other input in regards to updates. A base prototype will be developed and then expanded upon throughout the lifecycle of the app. This process will cycle through gathering requirements, designing and constructing new code around them, testing the updated prototype, releasing it, and receiving feedback. Each step in the process will be visited multiple times throughout development.

2.1.1 Requirement Gathering

a) **Activities:** Sourcing information on how the app should function, as well as any additional feedback received in previous iterations of the development process.

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b) **Entry Criteria:** Any further information regarding the prototype that has yet to be documented or implemented.
c) **Exit Criteria:** A list of updated requirements regarding what functionality, appearance, or other criteria the prototype should have.

2.1.2 Project Development

a) **Activities:** Creating code for various requirements established in the previous step. Potentially adding basic functionality, additional functionalities, or GUI elements to enhance the app.
b) **Entry Criteria:** A list of updated requirements regarding what functionality, appearance, or other criteria the prototype should have.
c) **Exit Criteria:** Finalized code for all current requirements.

2.1.3 Project Testing

a) **Activities:** Creating and rerunning prior tests for both updated features and prior work.
b) **Entry Criteria:** Finalized code for all current requirements.
c) **Exit Criteria:** Prototype passing all tests developed in this phase.

2.1.4 Project Release

a) **Activities:** Pushing the updated code to a public venue, as well as updating any user manuals or relevant information for running the current prototype.
b) **Entry Criteria:** Prototype passing all tests.
c) **Exit Criteria:** Submission of current prototype and all accompanying documentation.

2.1.5 Project Feedback

a) **Activities:** Receiving information on project performance once submitted as a tested prototype.
b) **Entry Criteria:** Submission of current prototype and all accompanying documentation.
c) **Exit Criteria:** Outside commentary, remarks, or other feedback on prototype functionality and success.

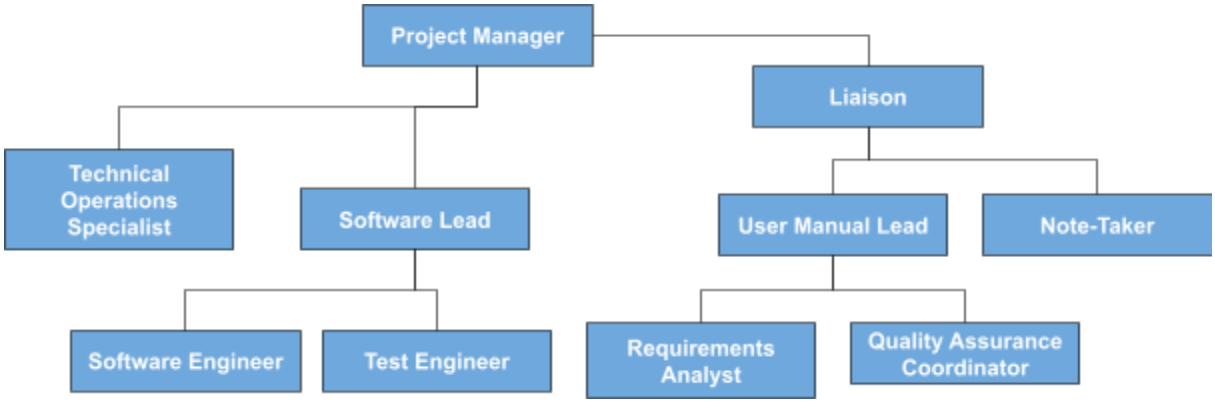
2.2. Organizational Structure

The project will have a primary point of contact between the instructor and the team through one of the participating students, and otherwise will have decisions dictated by relevant areas of expertise. The Project Manager will get final say on which direction documentation is taken in, as well as what order in which updates to the code or writing will be implemented.

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The student creating the repository will be the team's Software Lead and will be the only member of the team in charge of merging updated code to the main branch. Similarly, document submission will be done through the Liaison, requirements lists will be updated and maintained on the repository by the Requirements Analyst, and any tests will be performed by a Test Engineer.

Communication will be recorded and upheld through both in-person meetings as well as online communication tools such as Discord. Members of the team will be held accountable for their work by the Liaison, as the point of contact.



2.3. Roles and Responsibilities

On the team, each individual will fulfill multiple roles, as well as a handful of overlapping responsibilities. Both team members will contribute to the documentation, code, and live testing regarding the project.

Name	Roles
Denny Huang	Project Manager, Note-Taker, Software Lead, Requirements Analyst, Technical Operations Specialist
JD Faber	Liaison, Systems Engineer, Test Engineer, Quality Assurance Coordinator, Subject Matter Expert, User Manual Lead

3. Managerial Process Plans

3.1. Management objectives and priorities

Roles and Responsibilities

- Team Management: Organize meetings, make sure to keep good communication, and drive results.
- Liaison: Maintain contact with both team members, report relevant questions to the instructor, and submit assignments and deliverables.

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- Software Development: Maintain git repo, use feature branches, and keep main clean.
- Documentation Management: Create and upkeep documentation, and proofread.

Team-wide Priorities

- Prioritize accessibility features before working on UI.
- Ensure on-time submission for all deliverables.
- Properly review work before submission.

3.2. Assumptions, dependencies, and constraints

The team operates under a set core assumptions and constraints that decide how decisions are made and how responsibilities are carried out.

The team assumes that, while a primary point of contact has been designated, collaboration will proceed and any conflicts, whether interpersonal or not, will remain minimal and will not escalate to the point of requiring intervention or a unilateral decision by any members. Should disagreements arise, the team will resolve it through open discussion and mutual compromise between each other.

The primary constraint affecting the management structure of this project is the size of the team. As a two-person team, distribution of management roles cannot be perfectly done. Because of that, management responsibilities are more informally distributed, and decision making will rely on direct communication and consensus between two members rather than a delegation of multiple members.

3.3. Risk management

Regarding the code base, the risks surrounding the code will be dealt with by both team members hosting some form of local code that they can edit and run on their personal machines, as well as the updated code within the repository. Branches will be made and will not be merged until thoroughly tested. Since the repository will have some form of version history, any corrupted or non-functional code accidentally pushed through can be reverted to a previous state.

Documentation will be kept both on the repository and in a Google Drive. All documentation will also have a version history to revert deleted or poorly modified text, as well as a downloaded copy on the repository in case any online document gets corrupted at the source. Any and all changes will initially be made on Google Drive before exporting and saving onto the repository.

Risk management tasks will be completed throughout the project lifecycle, with key dates being deliverable deadlines. By incorporating risk management into the team's activities, there is only an estimated 5-10% increase in the project effort, or by a few additional hours unless a major risk is defined.

Task	Roles	Target Date

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Risk Identification	Everyone	10/09/2026
Risk Prioritization	Project Lead	10/12/2026
Risk Mitigation	Software Lead, Software Engineer, Test Engineer	10/18/2026
Risk Resolution	Project Lead	10/18/2026

3.3.1 Risk Identification and Analysis

Throughout the course of the project, regular meetings will be held where the team will discuss any updates made to the code, requirements, or other aspects of the project. During these meetings any changes will be discussed in depth, including discussion of affected portions of the project from any change. If any portion of the project seems to be majorly impacted by a new addition, the team will determine how much the impact will affect functionality or speed of development, and how likely it is that this risk would occur.

3.3.2 Risk Prioritizing

When risks are found to exist, the Project Manager will determine which risks will be most likely to happen or which risks will have the largest impact on the project and can delegate team members to work on them accordingly. Any risk that is both likely and highly impactful will be prioritized over risks that are unlikely and non-impactful. The likelihood of the risk will outweigh the impact in the team's prioritization.

4. Technical Process Plans

4.1. Methods, tools, and techniques

4.1.1 Software Tools and Development Methodology

We will be using Google Docs to create and collaborate on documentation. We will also be using Android Studio which simulates an android OS on PC. We aim to use YOLOv8 for object detection, utilizing VSCode and Android Studio, as well as some python for integrating YOLOv8. We will also be using GitHub for version control.

4.1.2 Quality Assurance Practices

To assure quality of the project, the team will be doing code review of any pull requests done on the repository as well as different tests on the code itself, such as using unit tests for various input regarding the routes. Additional quality assurance will be done with VSCode's debugging features,

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integration tests, and rerunning the prototype when various features are added to the project. We will also be using document QA practices, utilizing checklist reviews against the rubric, peer proofreading, and traceability verification between WRS sections.

4.2. Software documentation

The primary documentation created for this project will be regarding the project deliverables and repository development. Documentation will fall into one of these two categories for the most part, with a WRS document, user manuals, prototype slides, and a README and list of updates for the repository. Additional documentation will be added as-needed based on the team's discretion. All documents will be stored both in the repository itself and a separate Google Drive.

Document	Template or Standard	Created By	Reviewed By	Target Date	Distribution
WRS Document	484 - WRS.pdf	Denny	JD	10/09/2026	GitHub, Google Drive
User Manuals	PDF	JD	Denny	10/09/2026	GitHub, Google Drive
Mockup Sketches	484 - WRS.pdf	JD	Denny	10/09/2026	GitHub, Google Drive
AS-IS/TO-BE Slides	G-Slides	JD	Denny	10/16/2026	GitHub, Google Drive
Repository Update Notes	G-Docs	Denny	JD	10/16/2026	GitHub, Google Drive
Recorded Presentation	G-Docs	JD	Denny	10/16/2026	GitHub, Google Drive
AI Usage / Teamwork Report	G-Docs	Denny	JD	10/18/2026	GitHub, Google Drive

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